

I. SUPPLEMENTARY MATERIAL

A. The LLM-Guided Subgraph Extraction Prompts

Suggest GNN Features Prompt

- You are an expert in machine learning feature selection, specifically for the GNN graph machine tasks.
- Think about information required to accurately **<task>**.
- Return a numbered list of items without explanation.
- Sort the list according to item importance.

DBLP Example Output

1. Keywords/Topics extracted from the publication’s title and abstract
2. Venues of publications cited by this publication
3. Venues of previous publications by the authors of this publication
4. Abstract text or embeddings of the publication
5. Title text or embeddings of the publication
6. Research domains/sub-fields associated with the publication’s content
7. Co-author network’s historical venues
8. Common entities (e.g., specific algorithms, datasets, tools) mentioned in the publication
9. Average prestige/impact factor of venues where authors have previously published
10. Years of publication of cited works
11. Author’s affiliation type or research focus
12. Publication year of the current paper

Features to BGPs Mapping Prompt

- You are an expert in machine learning feature selection for graph machine learning tasks.
- The following describes the **<KG>** knowledge graph schema, detailing the relationships between graph entities in a series of triples, one triple per line:
<KG-schema>
- Given the following list of key features, select the matching relations from the previous schema.
<suggested-features>
- Think carefully and refine your selected/matching items Return the top **<K>** matched schema triples sorted by importance.
- Output only one selected triple per line without any explanation.

DBLP Schema BGP’s and their stats **<KG-schema>** as features to help LLM decide the query’s relevant BGP’s

subject , predicate , object , count
schema:Publication , schema:title , str , 6008680
schema:Publication , dblp:yearOfEvent , purl:ResourceIdentifier , 11404719
schema:Publication , dblp:publishedInJournalVolume , str , 10256566
schema:Publication , schema:bibtexType , str , 6008680
schema:Publication , rdfs:label , str , 6008680
schema:Publication , schema:authoredBy , str , 5808242
schema:Publication , schema:publishedInSeries , str , 3057949
schema:Publication , schema:numberOfCreators , str , 6008680
schema:Publication , schema:publishedBy , schema:Publication , 3057103
schema:Publication , schema:doi , str , 4842535
...

B. SPARQL generation and refinement prompts

BGPs To SPARQL Prompt

-You are an expert SPARQL query writer.

- Given the following triples list from the **<KG>** knowledge graph schema, write a SPARQL query to select the **<VT>** and its associated information given in the following triples list.
- The triples are directed; make sure to fulfill the direction and relation type.
- The query must return the union of sub-select statements in the form ?s ?p ?o.
- Each triple is Subject Entity - relation - Object Entity.
- Start with the **<VT>** node.

<BGP-List>

<SPARQL-Example>

_____Rules_____

- 1- Write nested select sub-queries and Union them.
- 2- In single-hop nested select, make sure to start the first BGP with the variable ?s.
- 3- In tow-hop or more nested select:
 - 3.1 Start the first BGP with the variable ?s, then use other variable names for next BGPs.
 - 3.2 Use the last connected entity as the subject, as shown in the previous example.
- 4- Generate only the SPARQL query without any explanation.
- 5- Make sure to use each given BGP triple.
- 6- add the BGP: 'Values ?s **<VT-List>**.' to the end of each sub query.
- 7- Refine all rules and the query syntax. 8- Do invent new relations i.e, dblp:authoredBy can not be dblp:Authored, But you can start with ?o instead of ?s.

Example:

?s a dblp:Publication.

?s dblp:authoredBy ?o.

_____ Should Be _____

?o a dblp:author. ?s dblp:authoredBy ?o.

SPARQL Refine Prompt

-You are an expert SPARQL query writer. Given the following SPARQL query, re-write it to follow the following rules.

- Rule1: Keep the nested selects and their Union statements.
- Rule2: restructure the n-hop sub-select to choose the latest BGP subject and object and as the select items.
- Example: { ?s a prefix:x. ?s prefix:y ?y. ?y prefix:z ?z. }

the latest BGP is ?y prefix:z ?z, then the select items must be: 1- ?y as ?s. 2- ?p. 3- ?z as ?o

_____ SPARQL Query _____

<sparql-query>

- Refine The Rules and Examples Carefully.
- Return only the Query; do not return any explanation.

<Answer>

An example of LLM-Guided SPARQL Query generated for the DBLP-PV NC task.

```
PREFIX dblp: <https://dblp.org/rdf/schema>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?s ?p ?o
FROM <https://www.dblp.org>
WHERE {
  { SELECT ?s ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:title ?o. BIND( "dblp:title" AS ?p). VALUES ?s {<VT-List>}. }}
  UNION
  { SELECT ?s ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:yearOfEvent ?o. BIND( "dblp:yearOfEvent" AS ?p). VALUES ?s {<VT-List>}. }}
  UNION
  { SELECT ?s ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:publishedInJournalVolume ?o. BIND( "dblp:publishedInJournalVolume" AS ?p). VALUES ?s {<VT-List>}. }}
  UNION
  { SELECT ?author ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:authoredBy ?author. ?author dblp:primaryAffiliation ?o. BIND( "dblp:primaryAffiliation" AS ?p). VALUES ?s {<VT-List>}. }}
  UNION
  { SELECT ?author ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:authoredBy ?author. ?author rdfs:label ?o. BIND( "rdfs:label" AS ?p). VALUES ?s {<VT-List>}. }}
  UNION
  { SELECT ?s ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:publishedInSeries ?o. BIND( "dblp:publishedInSeries" AS ?p). VALUES ?s {<VT-List>}. }}
  UNION
  { SELECT ?s ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:numberOfCreators ?o. BIND( "dblp:numberOfCreators" AS ?p). VALUES ?s {<VT-List>}. }}
  UNION
  { SELECT ?s ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:publishedBy ?o. BIND( "dblp:publishedBy" AS ?p). VALUES ?s {<VT-List>}. }}
  UNION
  { SELECT ?s ?p ?o WHERE { ?s a dblp:Publication. ?s dblp:doi ?o. BIND( "dblp:doi" AS ?p). VALUES ?s {<VT-List>}. }}
}
```

C. KG-WISE performance while varying the inference query size

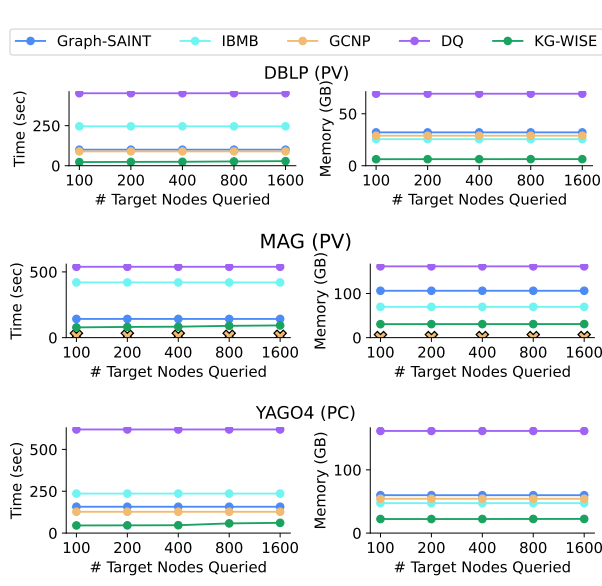


Fig. 1. Node classification inference time and memory usage of KG-WISE compared to SOTA GNN accelerators as the number of target nodes per query increases. The top, middle, and bottom rows correspond to the DBLP, MAG, and YAGO NC tasks, respectively. KG-WISE exhibits superior performance in all cases.

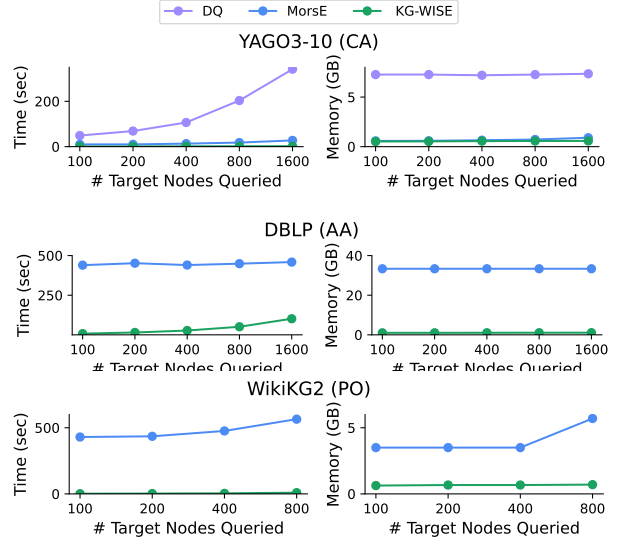


Fig. 2. Link prediction inference time and memory usage of KG-WISE vs. SOTA GNN accelerators across varying numbers of target nodes per query. The top, middle, and bottom rows correspond to the YAGO3-10-CA, DBLP-PV, and WikiKG-PO LP tasks, respectively. Baseline methods show near-linear inference time and constant memory usage, regardless of query size. In contrast, KG-WISE exhibits sub-linear growth in both time and memory, enabling efficient, query-aware, on-demand inference.

D. KG-WISE training pipeline v.s. existing GNN training and inference accelerators

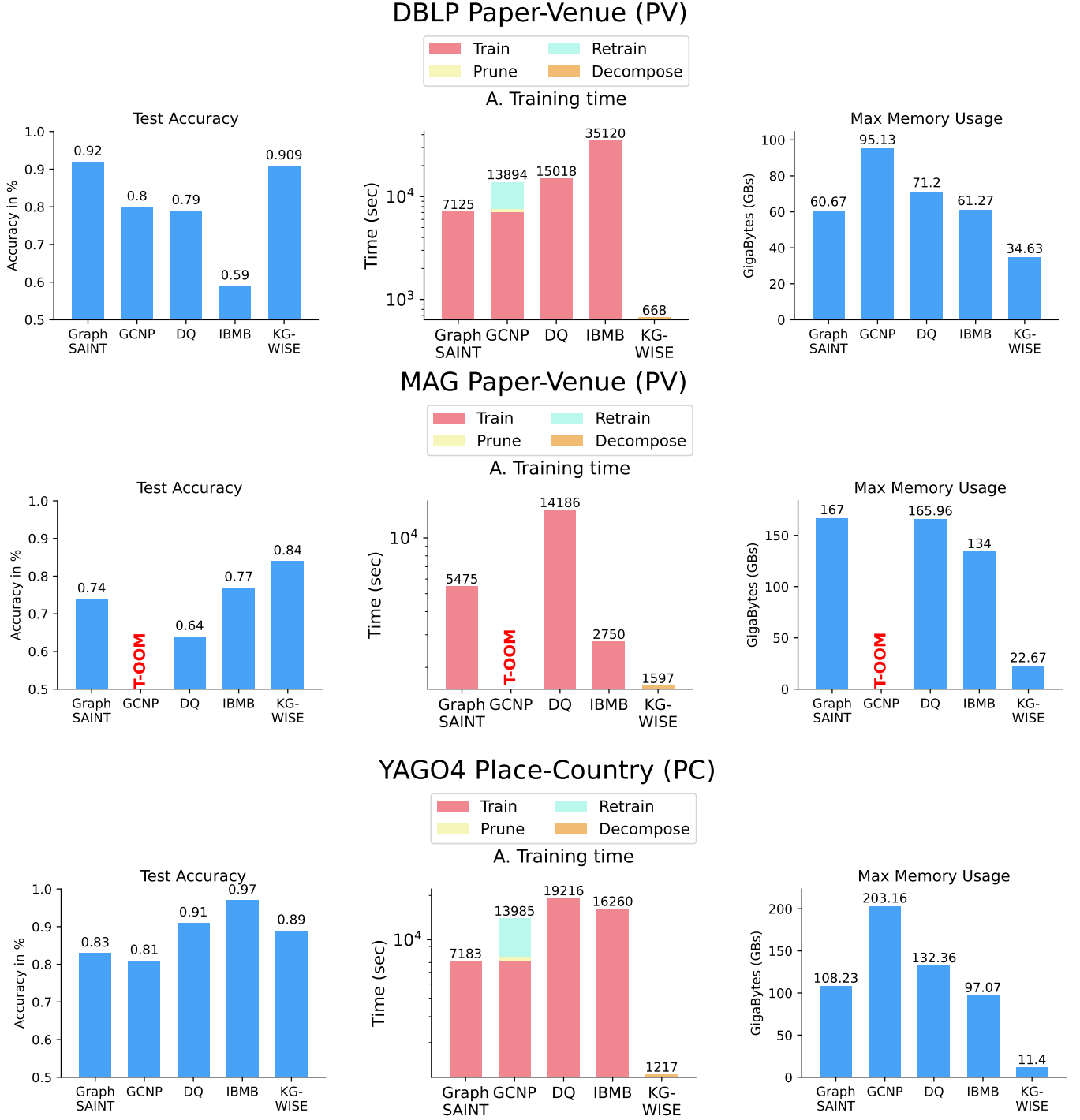


Fig. 3. Training metrics on NC tasks is measured using three metrics: (A) Test Accuracy (higher is better), (B) Training Time (lower is better), and (C) Max Memory Usage (lower is better). The top and middle sections show results for the Paper-Venue task on DBLP and MAG, respectively, while the bottom presents results for the Place-Country task on YAGO4. KG-WISE achieves comparable accuracy to SOTA methods and outperforms them with up to 12x faster inference and 91% memory reduction on YAGO4.